

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	1 of 10

Document Title

Inspection and Test Plan for 400kV Power Cables and Accessories

Project Name

GTC/1288/2025

0	1-June-2026	Issued For Bidding	J.M.Kim	Y.W.Park	S.T.Park	
Rev. No.	Date	Descriptions	Prepared By	Checked By	Approved By	Approved By
			LS Cable & System Ltd.			Client



	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	2 of 10

Table of Contents

1.	Introduction	3
1.1.	Purpose of Document	3
1.2.	Abbreviations and Definitions	3
1.2.1.	Abbreviations and Acronyms	3
1.2.2.	Definitions	3
1.3.	References	4
1.3.1.	International Codes, Standards, Recommendations, Specifications and Documents	4
2.	Inspection and Test Plan	5
2.1.	Test Plan for 400kV XLPE Power cable and accessories	5
2.1.1.	Routine Test (400kV Power cable)	5
2.1.2.	Additional Regular Test (400kV Power cable)	6
2.1.3.	Sample Test (400kV Power cable)	8
2.1.4.	Routine Test (Joint)	8
2.1.5.	Sample Test (Joint)	8
2.1.6.	Additional Test (Joint)	9
2.1.7.	Routine Test (Termination)	9
2.1.8.	Sample Test (Termination)	9
2.1.9.	Routine Test (Link Box and Sheath Voltage Limiters)	10

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	3 of 10

1. Introduction

1.1. Purpose of Document

This inspection and test plan covers testing procedure for AC 400kV power cable and accessories not including bonding/earthing cable, surge arrestors/sheath voltage limiter and Middle Voltage.

1.2. Abbreviations and Definitions

1.2.1. Abbreviations and Acronyms

Table1- 1 Abbreviations

Abbreviations	Definitions	Remarks
IEC	International Electrotechnical Commission	
H	HOLD POINT (H): Witnessing of the specified test or control by Inspector is mandatory. The specified test and control cannot be carried out in absence of inspector indicated.	
W	WITNESS POINT (W): Witnessing of the specified test or control by Inspector is not mandatory, but optional. Records of the inspections, tests and control conducted must be available to all the Inspector for review.	
R	REVIEW (R): Required records, reports or certificates of the controls and tests conducted must be available for review. Said documentation shall be made available immediately upon conclusion of the specified test or control.	
(W)	Spot Witness	
M	Monitoring	
AC	Alternating Current	
DC	Direct Current	
FAT	Factory Acceptance Test	
IEC	International Electro-technical Commission	
PE	Polyethylene	
XLPE	Cross-Linked Polyethylene	
L/B	Link Box	
MRB	Manufacturing Record Book	
IRC	Inspection Release Certificate	
NCR	Non-Conformance Report	
PD	Partial Discharge	

1.2.2. Definitions

Table1- 2 Definitions

Terms	Definitions	Remarks
Client	KAHRAMAA	KM
Vendor	LS Cable & System	LSCNS
Projects	GTC/1288/2025	

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	4 of 10

1.3. References

1.3.1. International Codes, Standards, Recommendations, Specifications and Documents

Table 1-3 International Codes, Standards and Recommendations

Document Number	Document Title
IEC 62067: 2022	Power cables with extruded insulation and their accessories for rated voltages above 150 kV ($U_m = 170$ kV) up to 500 kV ($U_m = 550$ kV) – Test methods and requirements
IEC 60228: 2023	Conductors of insulated cables
IEC 60229: 2007	Electric cables – Tests on extruded oversheaths with a special protective function
IEC 60885-3: 2015	Electrical test methods for electric cables – Part 3: Test methods for partial discharge measurements on lengths of extruded power cables
IEC 60811-201: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 201: General tests – Measurement of insulation thickness
IEC 60811-202: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 202: General tests – Measurement of thickness of non-metallic sheath
IEC 60811-203: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 203: General tests – Measurement of overall dimensions
IEC 60811-403: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 403: Miscellaneous tests – Ozone resistance test on cross-linked compounds
IEC 60811-404: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 404: Miscellaneous tests – Mineral oil immersion tests for sheaths
IEC 60811-501: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 501: Mechanical tests – Tests for determining the mechanical properties of insulating and sheathing compounds
IEC 60811-507: 2012	Electric and optical fibre cables – Test methods for non-metallic materials – Part 507: Mechanical tests – Hot set test for cross-linked materials

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	5 of 10

2. Inspection and Test Plan

2.1. Test Plan for 400kV XLPE Power cable and accessories

2.1.1. Routine Test (400kV Power cable)

Table 2-1 Routine Test

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Voltage test	IEC 62067: 2022 C 9.3	AC 750kV Voltage tester	No breakdown at 440kV for 60min	LS: 100% FAT: 10% of Shipping drums	H	(W)
2)	Partial discharge test	IEC 62067: 2022 C 9.2	AC 750kV Voltage tester PD Calibrator	Max.3pC at 345kV		H	(W)
3)	Dielectric loss angle	IEC 62067: 2022 C 12.4.5	Tan delta tester	Max. 0.0005 at 345kV (1.5U ₀)		H	(W)
4)	Conductor examination • Check of no. of wires • Ratio of diameters of two different wires • Conductor diameter	IEC 62067:2022 C 10.4, IEC 60228	Thickness meter Vernier calipers	As per approved Data Sheet		H	(W)
5)	Measurement of electrical resistance of conductor	IEC 62067:2022 C 10.5, IEC 60228	Digital Low Resistance Ohmmeter	As per approved Data Sheet		H	(W)
6)	Oversheath voltage withstand test	IEC 62067: 2022 C 9.4	DC Voltage tester	No breakdown at DC 25kV for 1min		H	(W)

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	6 of 10

2.1.2. Additional Regular Test (400kV Power cable)

Table 2-2 Additional Regular Test

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Measurement of thickness of insulation, oversheath & metallic sheath	IEC 60811-201 IEC 62067: 2022 C 10.6	Thickness meter Projector	As per approved Data Sheet	LS: 100% FAT: 10% of Shipping drums	H	(W)
2)	Measurement of insulation concentricity	IEC 62067: 2022 C 10.6	Profile projector	Max 10%	LS: 100% FAT: 10% of Shipping drums	H	(W)
3)	Measurement of insulation purity	KM Spec. V3 S3.3.12.1 C1.4.4.8.5	Microscope	Void : Max. 0.02mm Number of Voids in Any Area of 10mm×10mm : Max. 10 Black and Metallic Particle : Max. 0.08mm	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
4)	Measurement of moisture content in extruded insulation and screens	KM Spec. V3 S3.3.12.1 C1.4.4.8.6	Moisture tester	Insulation :Max. 150ppm Screens : Max. 500ppm	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
5)	Hot set test for XLPE insulation	IEC 60811-507 IEC 62067: 2022 C 10.9	Aging Oven Recorder	Elongation under Load: Max. 175 %, Permanent Elongation after Cooling: Max. 15 %	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
6)	Insulation shrinkage test	IEC 60840:2020 C 12.5.16	Oven	Max. 4.5 %	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
7)	Measurement of resistivity of semiconducting screens	IEC 62067: 2022 C 12.4.11	Oven, Resistivity tester	Conductor Screen : Max. 1,000 Ω.m Insulation Screen : Max. 500 Ω.m	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
8)	Measurement of screen protrusions	KM Spec. V3 S3.3.12.1 C1.4.4.8.10	Hot oil chamber	There shall be No Screen Defects	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
9)	Measurement of capacitance	IEC 62067: 2022 C 10.10	Capacitance Meter	As per approved Data Sheet	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	7 of 10

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
10)	Measurement of crosslinking by product concentration in XLPE cables	KM Spec. V3 S3.3.12.1 C1.4.4.8.12	TGA Analyzer	a. Total weight change of test sample after 30min : Max. 1.25% of the original sample weight b. Rate of change in the first 5min : Max.0.14 % c. Average rate of change of sample weight between the Time of 15 – 30min.: Max. 0.01%	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
11)	FTIR	KM Spec. V3 S3.3.12.1 C1.4.4.8.13	FTIR Analyzer	Comply with the Results of Type Test.	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
12)	DSC	KM Spec. V3 S3.3.12.1 C1.4.4.8.13	DSC Analyzer	Comply with the Results of Type Test.	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
13)	TGA	KM Spec. V3 S3.3.12.1 C1.4.4.8.13	TGA Analyzer	Comply with the Results of Type Test.	LS: 20% of every extrusion run FAT: 10% of Shipping drums	H	(W)
14)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.1 C1.4.4.8.13	-	90 ~ 110µm / 100g: Max. 2 EA Bigger than 100µm / 100g: Max. 1 EA	On every batch of incoming material	R	R
15)	Smoothness measurement on the incoming semi-conducting material	KM Spec. V3 S3.3.12.1 C1.4.4.8.13	-	0.05 ~ 0.10mm / 200cm ² : Max. 10 EA 0.10 ~ 0.25mm / 200cm ² : Max. 5 EA Bigger than 0.26mm /200 cm ² : 0 EA	On every batch of incoming material	R	R
16)	Measurement of weight for copper and lead of cable construction	KM Spec. V3 S3.3.12.1 C1.4.4.8	ELECTRIC BALANCE	Copper: Approx 26.4kg/m Lead: Approx 14.6kg/m	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
17)	Measurement of density of HDPE & carbon content	KM Spec. V3 S3.3.12.1 C1.4.4.8	Density meter	Density : Min 0.94 g/cm ³ Carbon content : 2.5±0.5%	LS: 10% of every extrusion run FAT: 10% of Shipping drums	H	(W)
18)	Water penetration test	IEC 62067:2022 C12.5.14, Annex E, Annex F	Temperature recorder	No Water shall emerge from the ends of the test piece	One sample each 10 km of production length	H	(W)

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	8 of 10

2.1.3. Sample Test (400kV Power cable)

Table 2-3 Sample Test

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Power frequency voltage test of 5 U ₀ for 1 Hour	KM Spec. V3 S3.3.12.1 C1.4.4.8.14	Voltage tester	No breakdown occurs at A.C 1,150kV for 1 hour	One 20m sample from every 10 km of manufacturing length	H	(W)

2.1.4. Routine Test (Joint)

Table 2-4 Routine Test(Joint)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	HV withstand test at 2.5U ₀ for 30 minutes	KM Spec. V3 S3.3.12.1 C1.4.5.1	AC 600kV Voltage tester	No breakdown at 575kV for 30min	100% (LS) 10% sample (Witness)	H	(W)
2)	PD measurement at 1.5 U ₀	KM Spec. V3 S3.3.12.1 C1.4.5.1	AC 600kV Voltage tester PD Calibrator	Max. 5pC at 345kV (Background noise Max 2pC)		H	(W)
3)	X-Ray test	KM Spec. V3 S3.3.12.1 C1.4.5.1	X-Ray	Any Crack or Crack Development Shall Not Be Found.		H	(W)
4)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.1 C1.5	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	100% accessory (LS) 1 sample (Witness)	H	(W)

2.1.5. Sample Test (Joint)

Table 2-5 Sample Test(Joint)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.5.1	FTIR Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
2)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.5.1	-	90 ~ 110μm / 100g: Max. 2 EA Bigger than 100μm / 100g: Max. 1 EA	On every batch of incoming material	R	R
3)	DLA measurement of insulation at 90°C	KM Spec. V3 S3.3.12.2 C1.4.5.1	DLA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
4)	TGA	KM Spec. V3 S3.3.12.2 C1.4.5.1	TGA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	9 of 10

2.1.6. Additional Test (Joint)

Table 2-6 Additional Test(Joint)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	Mechanical stretch of each rubber component	KM Spec. V3 S3.3.12.1 C1.4.5.1.1	-	No damage	100% (LS) 10% sample (Witness)	H	(W)
2)	Resistance of outer semi-conducting screen	KM Spec. V3 S3.3.12.1 C1.4.5.1.1	Ohm meter	Max 20kΩ		R	R
3)	Integrity of insulated gap 25kV dc for 15 minutes	KM Spec. V3 S3.3.12.1 C1.4.5.1.1	DC Voltage tester	Shall withstand DC 25kV 15min		H	(W)

2.1.7. Routine Test (Termination)

Table 2-7 Routine Test(Termination)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	HV withstand test at 2.5U ₀ for 30 minutes	KM Spec. V3 S3.3.12.1 C1.4.5.1	AC 600kV Voltage tester	No breakdown at 575kV for 30min	100% (LS) 10% sample (Witness)	H	(W)
2)	PD measurement at 1.5 U ₀	KM Spec. V3 S3.3.12.1 C1.4.5.1	AC 600kV Voltage tester PD Calibrator	Max. 5pC at 345kV (Background noise Max 2pC)		H	(W)
3)	X-Ray test	KM Spec. V3 S3.3.12.1 C1.4.5.1	X-Ray	Any Crack or Crack Development Shall Not Be Found.		H	(W)
4)	Packing and Preservation Inspection	KM Spec. V3 S3.3.12.1 C1.5	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	100% accessory (LS) 1 sample (Witness)	H	(W)

2.1.8. Sample Test (Termination)

Table 2-8 Sample Test(Termination)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
1)	FTIR	KM Spec. V3 S3.3.12.2 C1.4.5.1	FTIR Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
2)	Cleanliness measurement on the incoming insulation material	KM Spec. V3 S3.3.12.2 C1.4.5.1	-	90 ~ 110μm / 100g: Max. 2 EA Bigger than 100μm / 100g: Max. 1 EA	On every batch of incoming material	R	R
3)	DLA measurement of insulation at 90°C	KM Spec. V3 S3.3.12.2 C1.4.5.1	DLA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)
4)	TGA	KM Spec. V3 S3.3.12.2 C1.4.5.1	TGA Analyzer	Comply with the Results of Type Tests	10% (LS) 10% sample (Witness)	H	(W)

	Inspection and Test Plan for 400kV Power Cables and Accessories	Ref. No.	-		
		Manufacturing location	228, Suchul-daero, Gumi-si Gyeongsangbuk-do, Korea		
Confidential and Proprietary Information	GTC/1288/2025	Client	KAHRAMAA	Rev. No.	0
		Date	1/June/2026	Page.	10 of 10

2.1.9. Routine Test (Link Box and Sheath Voltage Limiters)

Table 2-9 Routine Test(Link Box and Sheath Voltage Limiters)

NO	Test Items	Reference Standard /Document	Equipment	Acceptance Criteria	Sampling	Inspection Involvement	
						LS	KM
Link Box							
1)	Visual and dimensional inspection	KM Spec. V3 S3.3.12.6 C1.9.2	Measuring tape	Shall be meet with approved drawings	100% (LS) 10% sample (Witness)	H	(W)
2)	Insulation Resistance measurement test	KM Spec. V3 S3.3.12.6 C1.9.2 b	Insulation resistance measurement	Min. 10MΩ		H	(W)
3)	Contact resistance measurement test	KM Spec. V3 S3.3.12.6 C1.9.2 c	Contact resistance tester	Max. 10μΩ		H	(W)
4)	Water sealing test	KM Spec. V3 S3.3.12.6 C1.9.2 d	Water tank	No leakage of the air from the sample during the test.(0.3bar 15min)		H	(W)
5)	DC withstand test	KM Spec. V3 S3.3.12.6 C1.9.3 ii	DC Voltage tester	No flash over at DC 25kV 5mins		H	(W)
6)	Impulse voltage withstand test	KM Spec. V3 S3.3.12.6 C1.9.3 i	Impulse tester	No flash over ▪ Between links: ± 50kVp / 10 times. ▪ Between links and earth: ± 20kVp / 10 times		H	(W)
7)	Packing inspection	KM Spec. V3 S3.3.12.6 C1.10	-	The quantities of the products shall be satisfied with packing list and Preservation requirements	100% (LS) 1 sample per each type (Witness)	H	(W)
Sheath Voltage Limiters							
1)	Leakage Current Test at DC 5kV for 1 min.	KM Spec V3 S3.3.12.6 C1.9.1	DC Voltage Tester	Leakage Current – Max. 0.1mA	100% (LS) 10% of SVL (Witness)	H	(W)
2)	Insulation Resistance	KM Spec V3 S3.3.12.6 C1.9.1	Insulation resistance measurement	Min. 10MΩ		H	(W)